

Introduction to WSN and IoT-Enabling Technologies

Presenter: Hello friends, welcome to Engineering Panda Family. In this video, I'll explain wireless sensor network (WSN) technology to you. Before I explain what WSN technology is, let me give you a list of technologies enabling IoT systems. When it comes to IoT systems, there are various technologies like wireless sensor networks, cloud computing, big data analytics, communication protocols, and embedded systems. These technologies enable us to work with IoT systems. In this video, I'll discuss wireless sensor networks in great detail. Let us first see the basic architecture of a WSN. When it comes to architecture, the first major essential role is end nodes. You can observe multiple end nodes in this wireless sensor network. First, understand the meaning of these end nodes. These end nodes capture physical data using sensors. For example, to capture temperature, use a temperature sensor. To

End Nodes: Sensors and Wireless Communication

Presenter: measure soil moisture, use a moisture sensor. To detect gas, use a gas sensor. Sensors in end nodes measure physical data. Once sensed, the data is transmitted to a central system wirelessly, as per the name wireless sensor network. End nodes have sensors and communicate via wireless protocols. End nodes are embedded systems with a microcontroller interfacing multiple sensors. The microcontroller senses physical data and is attached to a Wi-Fi, Bluetooth, or Zigbee card. These allow end nodes to communicate to the central system. In this wireless sensor network, end nodes have sensors as well as

WSN Architecture: Data Flow to Gateway and User Applications

Presenter: embedded systems with microcontrollers and wireless protocol cards like Wi-Fi, Bluetooth, or Zigbee for wireless communication. Once physical data is measured, it is routed to the router wirelessly. The router receives data from multiple end nodes, passes it to the coordinator, which is the gateway. The gateway sends data to the internet, where a server stores all physical data. Users access it via applications on mobile or computer, showing data from all sensors like temperature, soil moisture, or air quality. This data reaches the application through end nodes.

WSN Basics: Nodes, Routers, and Coordinators

Presenter: End nodes use wireless protocols coordinated with routers, then transferred to the gateway. Via the gateway, data loads onto the server using the internet. That is our basic architecture. In WSNs, we have distributed devices with sensors monitoring environmental and physical conditions. WSNs consist of numerous end nodes, routers, and coordinators. End nodes are embedded systems measuring physical data, transferred to routers, then to coordinators. Multiple routers may be needed for large areas, but a single router suffices for small areas. End nodes have several sensors attached, measuring multiple physical quantities. Each

Flexibility in Routing and Roles in WSN

Presenter: end node can also act as a router since it is an embedded system, but it needs power. In low-end applications, dedicated routers are used where end nodes do not route. For small applications like a single room, an end node can act as a router. Routers route data packets from end nodes to the coordinator. The coordinator, or gateway, collects data from routers and transfers it to the internet. Wireless sensor networks use wireless protocols. Now let me explain the basic characteristics of wireless sensor networks. We use wireless protocols like Zigbee, Bluetooth, or Wi-Fi.

WSN Protocols, Range, Speed, and ISM Band

Presenter: Embedded modules used as end nodes have Zigbee, Bluetooth, or Wi-Fi cards. The microcontroller interfaces with sensors, senses data, and transfers it to the router using these protocols. The range is 10 to 100 meters. Zigbee, Bluetooth, and Wi-Fi operate within 10 to 100 meters. Speed ranges from a few hundred Hz to a few Mbps, sufficient for real-time physical data measurement. Operating frequency is in the ISM band—industrial, scientific, and medical—which is free to use without government permission or license, even in the USA. Usually, we use 2.4 GHz for Zigbee, Bluetooth, and Wi-Fi.

Low Power, Self-Organizing, and Robust Network Characteristics

Presenter: End nodes operate at low power, allowing battery operation without power concerns. WSNs are self-organizing networks, requiring no manual configuration. End nodes communicate directly to routers, which transfer to gateways, then to the internet, all per protocols. The network is robust because even if some nodes fail, the system reconfigures automatically using available nodes, avoiding single-point failure.

WSN Applications: Weather, Air Quality, and Soil Monitoring

Presenter: Now let me give examples of wireless sensor networks. Weather monitoring systems use WSNs to collect data like temperature and humidity, aggregate and analyze it for various applications. Indoor air quality monitoring uses multiple end nodes in rooms to monitor air quality and gas concentrations. Soil moisture monitoring uses WSNs at various locations, also measuring temperature and other parameters. Surveillance systems use WSNs for motion data. Smart grids use WSNs for monitoring at various points.

Farm Automation and Conclusion

Presenter: Many applications exist, like farms. For large farms, place end nodes at different locations to measure temperature, humidity, and soil moisture. Access data on the internet via mobile for control without human intervention. Wireless sensor networks are an essential part of IoT systems. I hope you have understood this. If anything you want to share, note it in the comments. I'll be happy to help. Thank you for watching this video.